

AI could use as much electricity as Argentina and Netherlands in the coming years !!!

Posted by u/____san____ • 0 points • 12 comments

This is chilling . Dude many villages near my city never had electricity at all. Hope companies figure out a solution around it. I have written a newsletter on this topic. You can check it out here 📌

<https://dailydialogue.substack.com/p/ais-electric-appetite>

Comments

u/Infamous-Tax-9131 • 1 points • 2023-10-19 16:15 UTC

not me just now realizing that AI needs electricity to operate 🤖

u/spencemode • 0 points • 2023-10-13 13:12 UTC

I'm cool with it

u/Apprehensive-Basis70 • 3 points • 2023-10-13 10:30 UTC

Nvidia's new H100 is 12× more power efficient than current models. It will go up for sale in Q2 2024. This will continue for a long long time.

AMD made similar claims about their next-gen models.

AI itself is going to become more efficient.

AI will also advance numerous technology fields that will inevitably balance it's energy use.

Hope this helps.

u/____san____ • 1 points • 2023-10-13 10:57 UTC

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>Hope this helps.

agreed

u/Apprehensive-Basis70 • 1 points • 2023-10-13 11:05 UTC

There are other cases as well that we haven't seen yet. An example of this is if AI can take over entire helpdesk, technical support, and customer service jobs then you could essentially close down hundreds of office buildings that are specifically for these things.

In doing so you absolutely have a lot of power use being centralized in AI, but you no longer have to power all of those office buildings, their equipment, break rooms, etc etc.

So while energy use will go up there are cases where it will also reduce at the same time, and the reduction will likely be more than the average energy use for this specific use case.

There are a lot of societal/ethical issues as well if AI takes over large amounts of jobs but since that's not the topic I'll avoid those and give another example;

All of those people who would be driving to their customer service jobs would no longer be commuting. Meaning their attribution to CO2 output will be much less than before. (Unless they get another job that is farther away)

Lots of questions, and not a ton of answers but I see AI as being a necessary evil. I may be impartial to AI though since I work in a National Laboratory in the US and work with AI every day, and therefore see it's potential to help.

u/Nicolas-Gatien • 2 points • 2023-10-13 03:35 UTC

Any chance you could link the specified study? Thanks!

u/___san___ • 0 points • 2023-10-13 03:44 UTC

Here you go 📌

(<http://The growing energy footprint of artificial intelligence - Cell Press>
[https://www.cell.com/joule/pdf/S2542-4351\(23\)00365-3.pdf](https://www.cell.com/joule/pdf/S2542-4351(23)00365-3.pdf))

u/Nicolas-Gatien • 1 points • 2023-10-13 03:54 UTC

Out of curiosity - where do you stand on this double-edged sword?

Do you think we should slow down Generative AI and focus on the environmental effects?

Do you think we should double down on Generative AI and use its power to circle back to the environmental effects after?

u/___san___ • 0 points • 2023-10-13 04:08 UTC

On one hand I think the capabilities of Generative AI are impressive and can be used for betterment like in the medical field. However, the environmental impact is a genuine concern. So, it's not necessarily a question of slowing down or doubling down, but finding a balanced

approach. We can't ignore the environmental effects, but we also shouldn't halt technological progress

u/Nicolas-Gatien • 1 points • 2023-10-13 05:33 UTC

What do you think that balanced approach might look like?

u/____san____ • 1 points • 2023-10-13 09:55 UTC

I am not an expert but here's what I think

We could scale the resources depending on the workload. This will help prevents over-provisioning and saves energy during periods of low demand

We could power AI operations with renewable energy sources, which can be more cost-effective in the long run and reduce the carbon emission

This what I think could be done but as I said I am no expert so some stalwart in this field could explain it better

What do you think we could do?

u/AutoModerator • 1 points • 2023-10-13 03:03 UTC

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